

Ziyao (Richard) Cui

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EDUCATION

Massachusetts Institute of Technology

Ph.D. in Operations Research

- Advisor: Rahul Mazumder

Cambridge, MA

August 2026 – Present

Duke University

B.S. in Computer Science & B.S. in Statistical Science (Minor in Economics)

- GPA: 4.0/4.0 (Summa Cum Laude)
- Advisors: Jian Pei, Cynthia Rudin

Durham, NC

August 2022 – May 2026

University of Oxford

Study Abroad Program in Mathematics and Economics

- Advisor: Seth Flaxman

Oxford, United Kingdom

January 2025 – June 2025

PUBLICATIONS

- **Ziyao Cui**, Minxing Zhang, and Jian Pei. 2026. On Membership Inference Attacks in Knowledge Distillation. *SIGKDD Explor. Newsl.* 28, 1 (June 2026), 32–40. <https://doi.org/10.1145/3820356.3820359>
- **Ziyao Cui** and Jian Pei. How Is Uncertainty Propagated in Knowledge Distillation? *arXiv preprint arXiv:2601.18909*, 2026.
- Sahoko Ishida, Mohammed Aheed Osman, **Ziyao Cui**, Uchenna Agu, Emily Becher, Gabriel Battcock, Daniel Hernandez, Duccio Piovani, Frances Knight, Seth Flaxman, and Kevin Tang. Mapping Subnational Vulnerability to Inadequate Micronutrient Intake using a Bayesian Small Area Estimation Framework. *arXiv preprint arXiv:2604.14971*, 2026. *Under review at the International Journal of Health Geographics.*
- **Ziyao Cui** and Edric Tam. Beyond the Laplacian: Interpolated Spectral Augmentation for Graph Neural Networks. *arXiv preprint arXiv:2511.11928*, 2025.
- **Ziyao Cui**, Minxing Zhang, and Jian Pei. Learning to Attack: Uncovering Privacy Risks in Sequential Data Releases. *arXiv preprint arXiv:2510.24807*, 2025.
- **Ziyao Cui**, Erick Jiang, Nicholas Sortisio, Haiyan Wang, Eric Chen, and Cynthia Rudin. Revisiting Broken Windows Theory. *arXiv preprint arXiv:2509.16490*, 2025. *Under review at AAAI 2027.*

HONORS AND AWARDS

NSF Graduate Research Fellowship	2026
GEM Fellowship (Declined)	2026
Alex Vasilos Memorial Award (Excellence in Computer Science Research), Duke University	2026
Graduation with Highest Distinction in Computer Science, Duke University	2026
Phi Beta Kappa Honor Society	2025
ASA Data Expo Challenge Professional Category, Honorable Mention	2024
Dean's List with Distinction, Duke University	2023-2026

SERVICES

Reviewer: *NeurIPS* (2025, 2026)

Teaching Assistant: Theory and Algorithms for Machine Learning (CS671), Fall 2024; Real Analysis I (MATH531), Spring 2025

INTERNSHIP EXPERIENCE

Outlier AI <i>Coding Expert and Reviewer</i>	May 2024 – Present <i>Remote</i>
Shanghai Research Institute of Computing Technology Co. <i>Research Product Development Intern</i>	May 2024 – August 2024 <i>Remote</i>

SKILLS

Languages: English, Mandarin, Spanish, German
Technical Skills: Python, PyTorch, L^AT_EX, Java, C, R, SQL, MATLAB
Professional Memberships: American Statistical Association (ASA), Royal Statistical Association (RSS),
Institute of Mathematical Statistics (IMS)